

# Yongho Shin

Postdoctoral researcher

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## RESEARCH INTERESTS

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Online/approximation/learning-augmented algorithms for combinatorial optimization problems

## EDUCATION

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**Ph.D. in Computer Science, Yonsei University, South Korea** Mar. 2018 – Aug. 2024

◊ Dissertation topic: Relaxing hard requirements of online optimization via learning augmentation and limited revocability

◊ Advisor: Hyung-Chan An

**B.S. in Computer Science, Yonsei University, South Korea** Mar. 2012 – Feb. 2018

◊ Awarded *high honors at graduation*

## EMPLOYMENT

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**Institute of Computer Science, University of Wrocław, Poland** Nov. 2024 – present

◊ Postdoctoral researcher

◊ Advisor: Jarosław Byrka

## RESEARCH PAPERS

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### *Published Work*

- [1] Changyeol Lee, Dahoon Lee, Jongseo Lee, **Yongho Shin**, and Changki Yun. Optimal learning-augmented algorithm for online bidding. *arXiv preprint arXiv:2605.07349*, 2026. **To appear in NeurIPS 2026.**
- [2] Fateme Abbasi, Martin Böhm, Jarosław Byrka, Matin Mohammadi, and **Yongho Shin**. Servicing matched client pairs with facilities. *arXiv preprint arXiv:2602.19680*, 2026. **Appeared in WAOA 2026.**
- [3] Jarosław Byrka and **Yongho Shin**. Online rounding schemes for edge cover. *arXiv preprint arXiv:2507.13159*, 2025. **Appeared in WAOA 2026.**
- [4] Fateme Abbasi, Hyung-Chan An, Jarosław Byrka, Changyeol Lee, and **Yongho Shin**. Approximating (weighted) chromatic correlation clustering via cluster LP. *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2026)*, 13:1-13:20, 2026.
- [5] **Yongho Shin** and Phanu Vajanopath. Parsimonious learning-augmented online metric matching. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, PMLR 306, 112669–112688, 2026.
- [6] Davin Choo, Billy Jin, and **Yongho Shin**. Learning-augmented online bipartite fractional matching. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 7671–7686, 2025.

- [7] Changyeol Lee, **Yongho Shin**, and Hyung-Chan An. Improved algorithms for overlapping and robust clustering of edge-colored hypergraphs: An LP-based combinatorial approach. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 70823–70855, 2025.
- [8] **Yongho Shin**, Changyeol Lee, Gukryeol Lee, and Hyung-Chan An. Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis. *ACM Transactions on Algorithms 22(1)*, 11:1–31, 2025. *Conference version in ICML 2023 [9]*.
- [9] **Yongho Shin**, Changyeol Lee, Gukryeol Lee, and Hyung-Chan An. Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis. In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, PMLR 202, 31539–31561, 2023.
- [10] Kangsan Kim, **Yongho Shin**, and Hyung-Chan An. Constant-factor approximation algorithms for parity-constrained facility location and  $k$ -center. *Algorithmica 85*, 1883–1911, 2023. *Conference version in ISAAC 2020 [12]*.
- [11] **Yongho Shin** and Hyung-Chan An. Making three out of two: Three-way online correlated selection. In *Proceedings of the 32nd International Symposium on Algorithms and Computation (ISAAC 2021)*, 49:1–49:17, 2021.
- [12] Kangsan Kim, **Yongho Shin**, and Hyung-Chan An. Constant-factor approximation algorithms for the parity-constrained facility location problem. In *Proceedings of the 31st International Symposium on Algorithms and Computation (ISAAC 2020)*, 21:1–21:17, 2020.

#### Preprints

- [13] Mateusz Basiak, Marcin Bieńkowski, **Yongho Shin**, and Agnieszka Tatarczuk. Pure tail constraints for online problems. *arXiv preprint arXiv:2609.35337*, 2026.
- [14] **Yongho Shin**, Changyeol Lee, and Hyung-Chan An. On optimal consistency-robustness trade-off for learning-augmented multi-option ski rental. *arXiv preprint arXiv:2312.02547*, 2023. *Merged to [8]*.
- [15] **Yongho Shin**, Kangsan Kim, Seungmin Lee, and Hyung-Chan An. Online graph matching problem with a worst-case reassignment budget. *arXiv preprint arXiv:2003.05175*, 2020.

## TEACHING

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University of Wrocław, Poland

◊ Algorithms with predictions

Summer 2026

Teaching assistant, Yonsei University, South Korea

◊ CSI2103/CCO2103 Data structures

Spring 2018 – 2021, 2023, 2024

◊ CSI3108 Algorithm analysis

Fall 2018 – 2021, 2023

◊ AIC2130 Computer algorithms for AI applications

Fall 2023

◊ GEK6205 Design and analysis of optimization algorithms

Fall 2023

## AWARDS

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High honors at graduation, Yonsei University

Feb. 2018

## ACADEMIC SERVICE

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◊ Reviewer, WAOA 2026

◊ Reviewer, Discrete Optimization, 2026

◊ Reviewer, NeurIPS 2026

- ◇ Reviewer, MFCS 2026
- ◇ Reviewer, IWOCA 2026
- ◇ Reviewer, ICML 2026
  - Gold reviewer (top 25%)
- ◇ Reviewer, Acta Informatica, 2026
- ◇ Program committee member, EC 2026
- ◇ Reviewer, STOC 2026
- ◇ Reviewer, APPROX 2025
- ◇ Reviewer, ICALP 2025
- ◇ Reviewer, Mathematical Programming, 2024
- ◇ Reviewer, WAOA 2024

## TALKS AND PRESENTATIONS

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| <p><i>New Frontiers in Learning-Augmented Algorithms</i>, Berlin, Germany</p> <ul style="list-style-type: none"> <li>◇ Title: Learning-augmented online bipartite fractional matching</li> <li>◇ Contributed talk</li> </ul>                                       | <p>Aug. 2026</p> |
| <p><i>APPROX 2026</i>, Boston, MA, USA</p> <ul style="list-style-type: none"> <li>◇ Title: Approximating (weighted) chromatic correlation clustering via cluster LP</li> </ul>                                                                                     | <p>Aug. 2026</p> |
| <p><i>ALPS@IJCAI</i>, Bremen, Germany</p> <ul style="list-style-type: none"> <li>◇ Title: Optimal learning-augmented algorithm for online bidding</li> <li>◇ Contributed talk</li> </ul>                                                                           | <p>Aug. 2026</p> |
| <p><i>Combinatorial Algorithms Seminar</i>, Inha University, Incheon, South Korea</p> <ul style="list-style-type: none"> <li>◇ Title: Primal-dual algorithms for edge-colored clustering</li> </ul>                                                                | <p>July 2026</p> |
| <p><i>ICML 2026</i>, Seoul, South Korea</p> <ul style="list-style-type: none"> <li>◇ Title: Parsimonious learning-augmented online metric matching</li> <li>◇ Poster presentation</li> </ul>                                                                       | <p>July 2026</p> |
| <p><i>HALG 2026</i>, Stockholm, Sweden</p> <ul style="list-style-type: none"> <li>◇ Title: Learning-augmented online bipartite fractional matching</li> <li>◇ Contributed talk and poster presentation</li> </ul>                                                  | <p>June 2026</p> |
| <p><i>SNU Math Seminar</i>, SNU, Seoul, South Korea</p> <ul style="list-style-type: none"> <li>◇ Title: Online rounding for set cover under subset arrivals</li> </ul>                                                                                             | <p>Jan. 2026</p> |
| <p><i>Combinatorics Seminar</i>, Hanyang University, Seoul, South Korea</p> <ul style="list-style-type: none"> <li>◇ Title: Primal-dual algorithms for edge-colored clustering</li> </ul>                                                                          | <p>Jan. 2026</p> |
| <p><i>SNU CSE Seminar</i>, SNU, Seoul, South Korea</p> <ul style="list-style-type: none"> <li>◇ Title: Learning-augmented online bipartite fractional matching</li> </ul>                                                                                          | <p>Dec. 2025</p> |
| <p><i>NeurIPS 2025</i>, San Diego, CA, USA</p> <ul style="list-style-type: none"> <li>◇ Title: Improved algorithms for overlapping and robust clustering of edge-colored hypergraphs: An LP-based combinatorial approach</li> <li>◇ Poster presentation</li> </ul> | <p>Dec. 2025</p> |

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| <i>NeurIPS 2025</i> , San Diego, CA, USA                                                                                       | Dec. 2025 |
| ◊ Title: Learning-augmented online bipartite fractional matching                                                               |           |
| ◊ Poster presentation                                                                                                          |           |
| <i>2025 Combinatorics Workshop</i> , IBS DIMAG, Daejeon, South Korea                                                           | Aug. 2025 |
| ◊ Title: Learning-augmented online bipartite fractional matching                                                               |           |
| ◊ Contributed talk                                                                                                             |           |
| <i>DGIST BRL AGSTA Seminar</i> , DGIST, Daegu, South Korea                                                                     | Aug. 2025 |
| ◊ Title: Online rounding for set cover under subset arrivals                                                                   |           |
| <i>Discrete Analysis Seminar</i> , Yonsei University, Seoul, South Korea                                                       | Aug. 2025 |
| ◊ Title: Online rounding for set cover under subset arrivals                                                                   |           |
| <i>Combinatorial Optimization Seminar</i> , Yonsei University, Seoul, South Korea                                              | Aug. 2025 |
| ◊ Title: Learning-augmented online bipartite fractional matching                                                               |           |
| <i>HALG 2025</i> , Zurich, Switzerland                                                                                         | June 2025 |
| ◊ Title: Learning-augmented algorithms for the multi-option ski rental problem                                                 |           |
| ◊ Contributed talk and poster presentation                                                                                     |           |
| <i>COG Seminar</i> , University of Wrocław, Wrocław, Poland                                                                    | May 2025  |
| ◊ Title: Learning-augmented online bipartite fractional matching                                                               |           |
| <i>COG Seminar</i> , University of Wrocław, Wrocław, Poland                                                                    | Dec. 2024 |
| ◊ Title: Online correlated selection                                                                                           |           |
| <i>Discrete Analysis Seminar</i> , Yonsei University, Seoul, South Korea                                                       | June 2024 |
| ◊ Title: Three-way online correlated selection                                                                                 |           |
| <i>Discrete Math Seminar</i> , IBS DIMAG, Daejeon, South Korea                                                                 | May 2024  |
| ◊ Title: Three-way online correlated selection                                                                                 |           |
| <i>ICML 2023</i> , Honolulu, HI, USA                                                                                           | July 2023 |
| ◊ Title: Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis |           |
| ◊ Poster presentation                                                                                                          |           |
| <i>Theory Tea</i> , Cornell University, Ithaca, NY, USA                                                                        | Dec. 2022 |
| ◊ Title: Three-way online correlated selection                                                                                 |           |
| <i>ISAAC 2021</i> , Fukuoka, Japan (virtual)                                                                                   | Dec. 2021 |
| ◊ Title: Making three out of two: Three-way online correlated selection                                                        |           |
| <i>AAAC 2021</i> , Tainan, Taiwan (virtual)                                                                                    | Oct. 2021 |
| ◊ Title: Making three out of two: Three-way online correlated selection                                                        |           |
| <i>ISAAC 2020</i> , Hong Kong, China (virtual)                                                                                 | Dec. 2020 |
| ◊ Title: Constant-factor approximation algorithms for the parity-constrained facility location problem                         |           |

## EXPERIENCE

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|---------------------------------------------------------------------------------------|-----------------------|
| <i>Co-organizer</i> of Yonsei CS theory student group, Yonsei University, South Korea | Jan. 2023 – Feb. 2024 |
| ◊ Initiated a reading group of TCS students in and out of Yonsei University           |                       |

◇ Organized seminar talks on various topics including mechanism design and quantum computing

*Intern*, Cornell University, USA

Sept. 2022 – Dec. 2022

◇ Director: David B. Shmoys

*Undergraduate intern*, Yonsei University, South Korea

Jan. 2017 – Feb. 2018

◇ Advisor: Hyung-Chan An

*Undergraduate voluntary tutor*, Yonsei University, South Korea

◇ CSI3108 Algorithm analysis

Fall 2016, 2017

◇ CSI2103 Data structures

Spring 2017

*Web programmer*, Republic of Korea Air Force, South Korea

Nov. 2013 – Aug. 2015

◇ In fulfillment of mandatory military service